



Conventional ML Methods

# Applied Machine Learning

Classifier Models :: Basic

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# Outline

#GeoAI #Mobility #Geography #STDS

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# Logistic Regression

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**Logistic Regression** is the most simplest and linear machine learning classifier among all models.

Although there is a word “**regression**” in the name, this method is used for binary categorization: yes and no, e.g., gender, binary levels, or disease confirmation.

It is a well-known and commonly used method in various fields, especially in medical and sociological studies.

Logistic regression conduct a non-linear function, **sigmoid**, to transform all values as a 0-to-1 scale probability, and we usually adopt 0.5 as a **threshold** for classification.

# Logistic Regression

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Suppose we have input features represented as a matrix:

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1m} \\ x_{21} & \cdots & x_{2m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{bmatrix}$$

The dependent variable is  $Y$  only a binary value.

$$Y = \begin{cases} 0 & \text{benign} \\ 1 & \text{malignant} \end{cases}$$

Here, we use the multi-linear function as the input variables  $X$ .

# Logistic Regression

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Here, we use the multi-linear function as the input variables  $X$ , like a linear regression (e.g., OLS).

$$z = \sum_{i=1}^m w_i x_i + b = w \cdot X + b$$

where  $b$  is bias and  $w_i = [w_1, w_2, \dots, w_m]$  is the  $i$ th weight or coefficient.

For a classification model, we need to bound the  $y$  from 0 to 1, then

$$P(Y = 1|X) = \sigma(z) = \frac{1}{1 + e^{-z}}, \text{ since } 0 \leq P(Y = 1|X) \leq 1$$

# Logistic Regression

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## Sigmoid Function

The sigmoid function converts the continuous variable data into the probability between 0 and 1.

$\sigma(z)$  tends toward to 1 as  $z \rightarrow \infty$

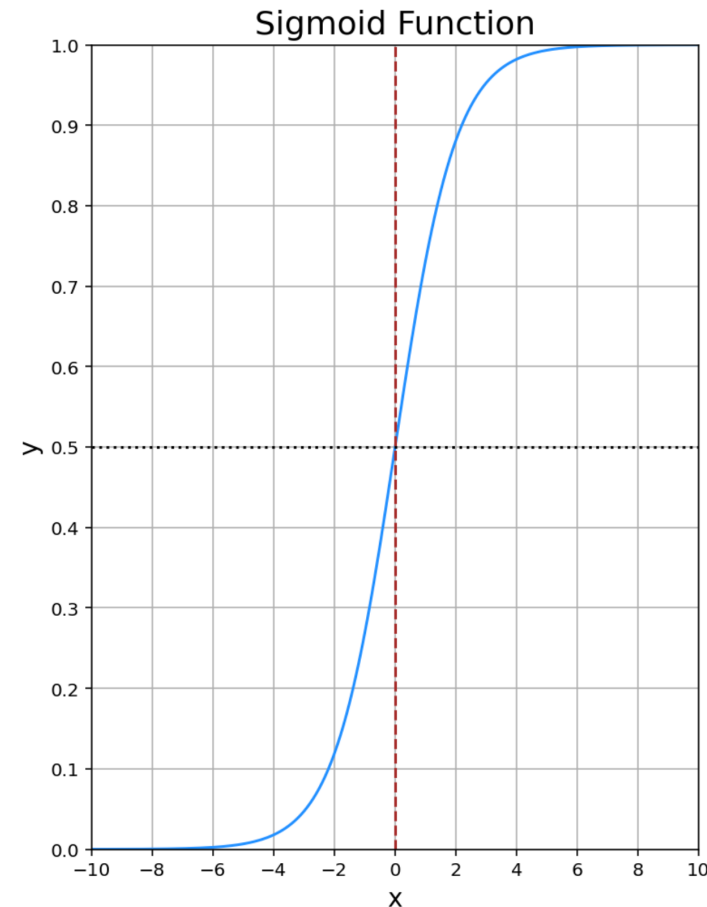
$\sigma(z)$  tends toward to 0 as  $z \rightarrow -\infty$

$\sigma(z)$  is bounded between 0 and 1

The probability of each class can be:

$$P(y = 1) = \sigma(z)$$

$$P(y = 0) = 1 - \sigma(z)$$



# K-Nearest Neighbors

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## K-Nearest Neighbors

What if the relationship between observations cannot be represented well by a simple linear decision boundary?

Here, we introduce another method, called K-Nearest Neighbor Classifier, conducted by identifying the K closest data points to a given input and making predictions based on the majority class or average value of those neighbors.

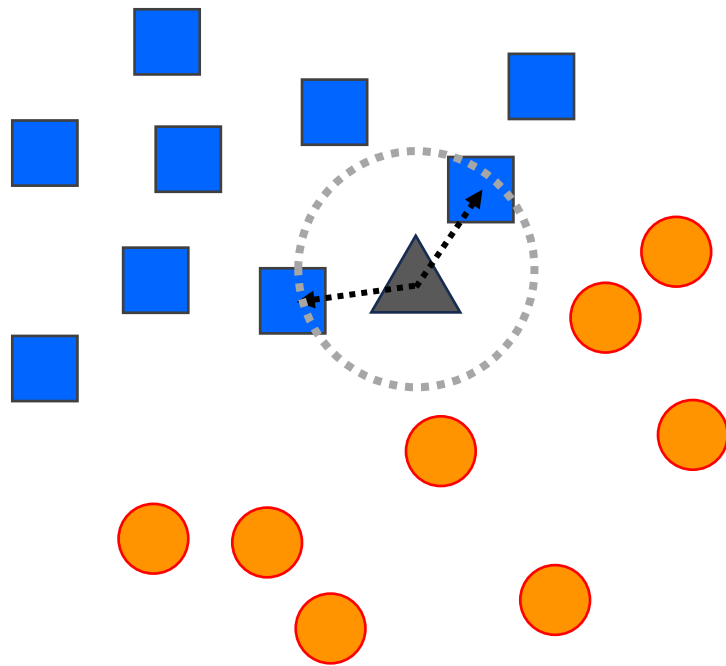
As a result, number of nearest neighbors is the key hyperparameters to significantly affect the model outcome.

→ Similar observations tend to have similar outcomes.

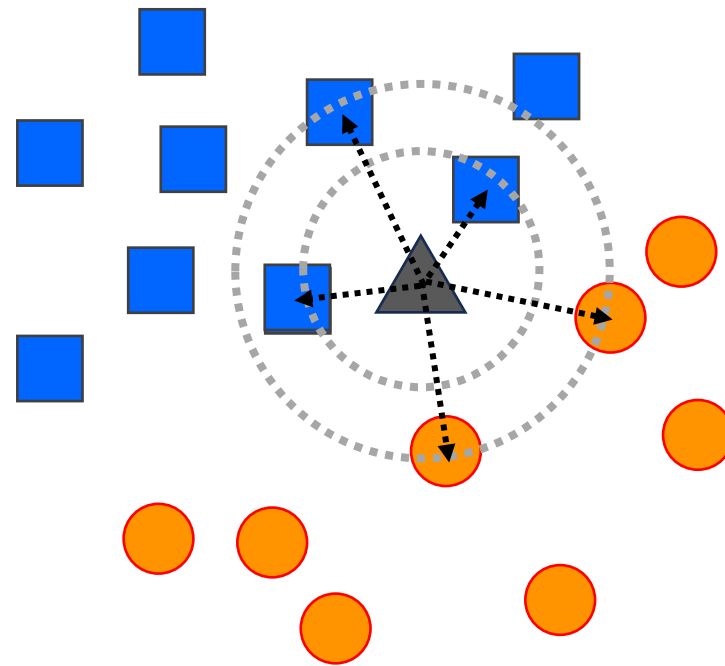
# K-Nearest Neighbors

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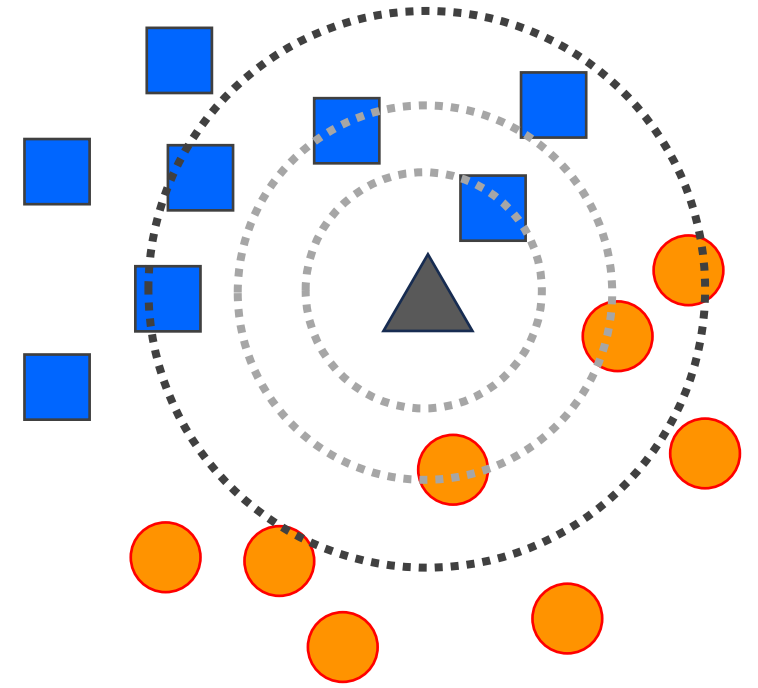
Here, we give three examples ...  $K=2, 5, 9$



2 Squares  $\rightarrow$  Square



4 Squares + 2 Circles  $\rightarrow$  Square



5 Squares + 4 Circles  $\rightarrow$  Square

# K-Nearest Neighbors

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## Hyperparameters

**n\_neighbors:** the number of nearest neighbors

**weights:** uniform and distance

**algorithm:** auto, ball\_tree, kd\_tree, brute

**leaf\_size:** Leaf size passed to BallTree or KDTree

**p:** Power parameter for the Minkowski metric. When  $p = 1$ , this is equivalent to using manhattan\_distance (L1), and euclidean\_distance (L2) for  $p = 2$ . For arbitrary  $p$ , minkowski\_distance ( $L_p$ ) is used.

# K-Nearest Neighbors

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## Weights

Weight function used in prediction. Possible values:

**uniform:** uniform weights. All points in each neighborhood are weighted equally.

**distance:** weight points by the inverse of their distance. in this case, closer neighbors of a query point will have a greater influence than neighbors which are further away.

# K-Nearest Neighbors

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## Minkowski metric

Metric to use for distance computation. Default is minkowski, which results in the standard Euclidean distance when  $p = 2$ .

$$d(a, b) = \left( \sum_{i=1}^n |a_i - b_i|^p \right)^{\frac{1}{p}}$$

**p=1** Manhattan Distance

**p=2** Euclidean Distance

**p=∞** Chebyshev Distance

# K-Nearest Neighbors

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## Algorithms

It determines how the algorithm finds the nearest data points during query time.

**Brute Force:** Computes the distance from the query point to **every single** sample in the training dataset and sorts them to find the top  $k$ .

**kd\_tree:** Recursively splits feature space along axis-aligned **hyperplanes** to build a binary tree.

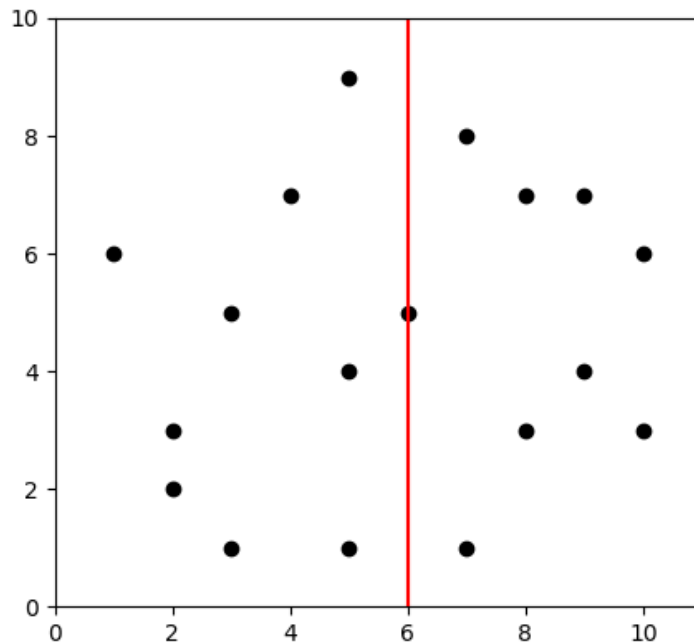
**ball\_tree:** Groups data points into nested  $D$ -dimensional **hyperspheres** (balls) defined by a centroid and radius.

# K-Nearest Neighbors

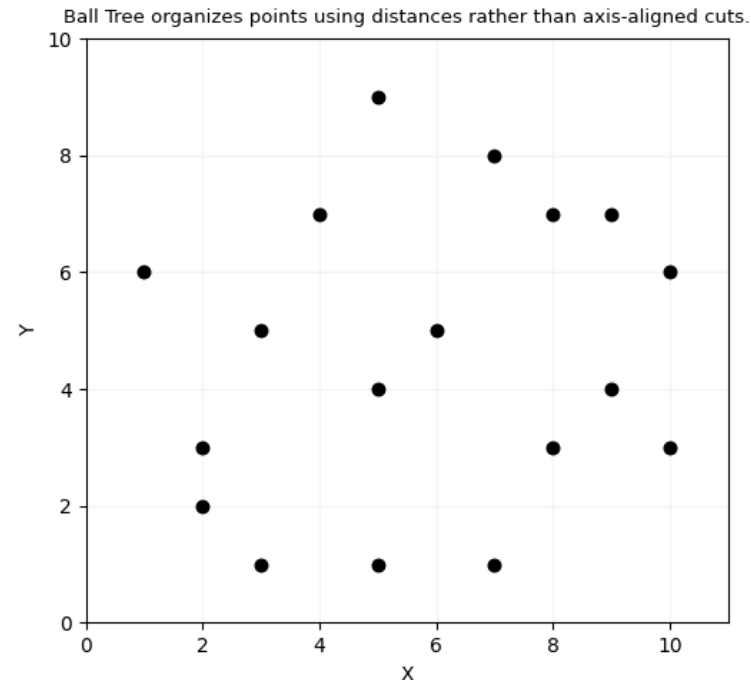
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## Algorithms KD Tree

Red = X-axis split  
Blue = Y-axis split



### 1. Original observations



## Ball Tree

Tree structure (top → bottom)

Root  
n=18

# K-Nearest Neighbors

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## Summary

KNN does **not learn** a formula or model during training, and **memorizes** the data and decides only during prediction.

We found that as dimensionality increases, the feature space grows exponentially, making the data increasingly sparse.

In high-dimensional spaces, finding enough **nearby** observations requires searching through a surprisingly large portion of the feature space → **Curse of Dimensionality**

Therefore, the differences in the relative distance of nearest and furthest observations could be much smaller.

# Naïve Bayes

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## Naïve Bayes

Instead of defining similarity geometrically, can we classify observations probabilistically?

Naive Bayes is a machine learning classification algorithm that predicts the category of a data point using probability. It assumes that all features are independent of each other.

# Naïve Bayes

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## Some Assumptions for Naïve Bayes Classifiers

### a) Feature Independence

Assumes every feature contributes to the classification independently, with no interactions or correlations between them.

### b) Normally Distributed Continuous Features

Assumes numeric, continuous data follows a Gaussian (normal) bell-curve distribution within each class.

### c) Multinomial Discrete Features

Assumes categorical or count data follows a multinomial distribution within each class.

# Naïve Bayes

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## Some Assumptions for Naïve Bayes Classifiers

### d) Equal Feature Weight

Assumes all features carry equal weight and importance when predicting the target class.

### e) No Missing Data

Assumes the dataset is complete, with no missing values.

# Naïve Bayes

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## Bayes' Theorem

It provides a principled way to reverse conditional probabilities.

$$P(y | X) = \frac{P(X | y) \cdot P(y)}{P(X)}$$

$P(y | X)$ : Posterior probability, probability of class  $y$  given features  $X$

$P(X | y)$ : Likelihood, probability of features  $X$  given class  $y$

$P(y)$ : Prior probability of class  $y$

$P(X)$ : Marginal likelihood or evidence

# Naïve Bayes

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The **naïve** in Naive Bayes comes from the assumption that all features are independent given the class.

$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) \cdot P(x_2 | y) \cdot \dots \cdot P(x_n | y)$$

Hence, Bayes' theorem becomes

$$P(y | x_1, x_2, \dots, x_n) = \frac{P(y) \cdot \prod_{i=1}^n P(x_i | y)}{P(X)}$$

Since the denominator is a constant for a given input, so

$$P(y | x_1, x_2, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n P(x_i | y)$$

# Naïve Bayes

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$$P(y \mid x_1, x_2, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n (x_i \mid y)$$

Thus, we compute the posterior for each class  $y$  and choose the class with

$$\hat{y} = \operatorname{argmax}_y P(y) \prod_{i=1}^n (x_i \mid y)$$

And this is Naïve Bayes Classifier.

# Naïve Bayes

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| Outlook  | Yes | No | P(Yes) | P(No) |
|----------|-----|----|--------|-------|
| Sunny    | 2   | 3  | 2/9    | 3/5   |
| Overcast | 4   | 0  | 4/9    | 0/5   |
| Rainy    | 3   | 2  | 3/9    | 2/5   |
| Total    | 9   | 5  | 9/9    | 5/5   |

| Temp. | Yes | No | P(Yes) | P(No) |
|-------|-----|----|--------|-------|
| Hot   | 2   | 2  | 2/9    | 2/5   |
| Mild  | 4   | 2  | 4/9    | 2/5   |
| Cool  | 3   | 1  | 3/9    | 1/5   |
| Total | 9   | 5  | 9/9    | 5/5   |

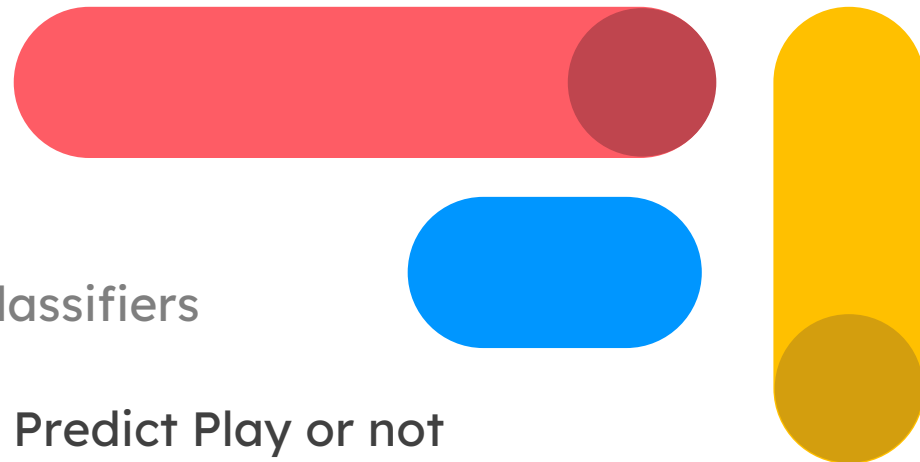
| Humidity | Yes | No | P(Yes) | P(No) |
|----------|-----|----|--------|-------|
| Hot      | 3   | 4  | 3/9    | 4/5   |
| Normal   | 6   | 1  | 6/9    | 1/5   |
| Total    | 9   | 5  | 9/9    | 5/5   |

| Wind  | Yes | No | P(Yes) | P(No) |
|-------|-----|----|--------|-------|
| False | 6   | 2  | 6/9    | 2/5   |
| True  | 3   | 3  | 3/9    | 3/5   |
| Total | 9   | 5  | 9/9    | 5/5   |

| Play | Yes | No | P(Yes) | P(No) |
|------|-----|----|--------|-------|
| Play | 9   | 5  | 9/14   | 5/14  |

# Naïve Bayes

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| Outlook      | P(Yes)     | P(No)      |
|--------------|------------|------------|
| <b>Sunny</b> | <b>2/9</b> | <b>3/5</b> |
| Overcast     | 4/9        | 0/5        |
| Rainy        | 3/9        | 2/5        |

| Temp.      | P(Yes)     | P(No)      |
|------------|------------|------------|
| <b>Hot</b> | <b>2/9</b> | <b>2/5</b> |
| Mild       | 4/9        | 2/5        |
| Cool       | 3/9        | 1/5        |

| Humidity      | P(Yes)     | P(No)      |
|---------------|------------|------------|
| Hot           | 3/9        | 4/5        |
| <b>Normal</b> | <b>6/9</b> | <b>1/5</b> |

| Wind         | P(Yes)     | P(No)      |
|--------------|------------|------------|
| <b>False</b> | <b>6/9</b> | <b>2/5</b> |
| True         | 3/9        | 3/5        |

| Play | P(Yes) | P(No) |
|------|--------|-------|
| Play | 9/14   | 5/14  |

Predict Play or not

Given  $X = \{\text{Sunny}, \text{Hot}, \text{Normal}, \text{False}\}$

For Play=Yes

$$P(\text{Yes} | \text{Today}) = \frac{2}{9} \times \frac{2}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14} = 0.0141$$

For Play=No

$$P(\text{No} | \text{Today}) = \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14} = 0.0069$$

Normalize the probability

$$P(\text{Yes} | \text{Today}) = \frac{0.0141}{0.0141 + 0.0069} = 0.673$$

$$P(\text{No} | \text{Today}) = \frac{0.0069}{0.0141 + 0.0069} = 0.327$$

As a result, the prediction will be **Play**

# Naïve Bayes

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Depends on posterior distribution, ...

|                      |                                                                          |
|----------------------|--------------------------------------------------------------------------|
| <b>BernoulliNB</b>   | Naive Bayes classifier for multivariate Bernoulli models.                |
| <b>CategoricalNB</b> | Naive Bayes classifier for categorical features.                         |
| <b>ComplementNB</b>  | The Complement Naive Bayes classifier described in Rennie et al. (2003). |
| <b>GaussianNB</b>    | Gaussian Naive Bayes (GaussianNB).                                       |
| <b>MultinomialNB</b> | Naive Bayes classifier for multinomial models.                           |

# Naïve Bayes

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## Advantages

- a) **Fast & Simple:** Easy to implement and computationally efficient.
- b) **Scalable & Data-Efficient:** Performs well with high-dimensional features and limited training data.
- c) **Handles Varied Data Types:** Works well with categorical data (and assumes normal distribution for continuous features).

# Naïve Bayes

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## Disadvantages

**Independence Assumption:** Assumes features are independent, which rarely holds in real-world data.

**Sensitivity to Noise:** Performance drops with irrelevant attributes.

**Zero-Frequency Problem:** Assigns zero probability to unseen categories, hurting generalization.

# Decision Tree

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## Decision Tree

Can we learn classification rules directly from the data without assuming a specific probability distribution or feature independence?

Decision Trees (DTs) are a non-parametric supervised learning method used for classification

For instance, in the example below, decision trees learn from data to approximate a sine curve with a set of if-then-else decision rules. The deeper the tree, the more complex the decision rules and the fitter the model.

# Decision Tree

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How do we determine which feature should be the first and values for splitting?

a) GINI index

$$gini = 1 - \sum_{i=1}^n p_i^2$$

b) Entropy

$$H(S) = - \sum_{i=1}^n p_i \log_2 p_i$$

| Urban       | Count | Prob  |
|-------------|-------|-------|
| Commerical  | 3     | 3/20  |
| Residential | 6     | 6/20  |
| Recreation  | 5     | 5/20  |
| Industrial  | 6     | 6/20  |
| Total       | 20    | 20/20 |

$$gini = 1 - \left[ \left( \frac{3}{20} \right)^2 + \left( \frac{6}{20} \right)^2 + \left( \frac{5}{20} \right)^2 + \left( \frac{6}{20} \right)^2 \right] = 1 - \frac{106}{400} = 0.735$$

$$H(S) = - \left[ \frac{3}{20} \log_2 \frac{3}{20} + \frac{6}{20} \log_2 \frac{6}{20} + \frac{5}{20} \log_2 \frac{5}{20} + \frac{6}{20} \log_2 \frac{6}{20} \right] = 1.953$$

# Decision Tree

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| ID | Age | Exercise | Disease |
|----|-----|----------|---------|
| A  | 25  | 6        | 0       |
| B  | 30  | 5        | 0       |
| C  | 35  | 4        | 0       |
| D  | 40  | 3        | 0       |
| E  | 45  | 5        | 0       |
| F  | 50  | 2        | 1       |
| G  | 55  | 4        | 0       |
| H  | 60  | 2        | 1       |
| I  | 65  | 1        | 1       |
| J  | 70  | 3        | 1       |

Number of samples ( $N$ ) is 10, where  $N_0 = 6$  and  $N_1 = 4$ .

Here, we use Gini index as an example

$$Gini_{root} = 1 - [p_0^2 + p_1^2] = 1 - \left[ \left( \frac{6}{10} \right)^2 + \left( \frac{4}{10} \right)^2 \right] = 0.48$$

Since  $Gini_{root}$  is not 0, we start with the following two features.

Let's start with Age and then Exercise.

# Decision Tree

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## Feature AGE

Values are (25, 30, 35, 40, 45, 50, 55, 60, 65, 70)

Using the mid-value between two neighboring values could be 27.5, 32.5, 37.5, ..., 67.5

Taking  $\text{Age} \leq 47.5$  as an example, we got

| ID | Age | Disease |
|----|-----|---------|
| A  | 25  | 0       |
| B  | 30  | 0       |
| C  | 35  | 0       |
| D  | 40  | 0       |
| E  | 45  | 0       |

$$\begin{aligned} N_L &= 5 & N_R &= 5 \\ Gini_L &= 1 - (1^2 + 0^2) = 0 & Gini_R &= 1 - (0.8^2 + 0.2^2) = 0.32 \\ Gini_{split} &= \frac{N_L}{N} Gini_L + \frac{N_R}{N} Gini_R = \frac{5}{10} \times 0 + \frac{5}{10} \times 0.32 = 0.16 \\ \text{Impurity reduction is} \\ \Delta Gini &= Gini_{root} - Gini_{split} = 0.48 - 0.16 = 0.32 \end{aligned}$$

| ID | Age | Disease |
|----|-----|---------|
| F  | 50  | 1       |
| G  | 55  | 0       |
| H  | 60  | 1       |
| I  | 65  | 1       |
| J  | 70  | 1       |

# Decision Tree

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## Feature Exercise

Taking  $Exercise \leq 2.5$  as an example, we got

| ID | Age | Disease |
|----|-----|---------|
| F  | 2   | 1       |
| H  | 2   | 1       |
| I  | 1   | 1       |

$$N_L = 3$$

$$N_R = 7$$

$$Gini_L = 1 - (1^2 + 0^2) = 0 \quad Gini_R = 1 - \left( \frac{1^2}{7} + \frac{6^2}{7} \right) = 0.245$$

$$Gini_{split} = \frac{N_L}{N} Gini_L + \frac{N_R}{N} Gini_R = \frac{3}{10} \times 0 + \frac{7}{10} \times 0.245 = 0.171$$

**Impurity reduction is**

$$\Delta Gini = Gini_{root} - Gini_{split} = 0.48 - 0.171 = 0.309$$

$$\Delta Gini_{age} > \Delta Gini_{exercise}$$

So, we choose age with  $Age \leq 47.5$  as the first split.

| ID | Age | Disease |
|----|-----|---------|
| A  | 6   | 0       |
| B  | 5   | 0       |
| C  | 4   | 0       |
| D  | 3   | 0       |
| E  | 5   | 0       |
| G  | 4   | 0       |
| J  | 3   | 1       |

# Decision Tree

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Therefore, we can formulize as ...

For each feature:  $X_1, X_2, \dots, X_n$ , then candidate splits are  $s_1, s_2, \dots$

We may compute Gini index for each split

$$I_{split} = \frac{N_L}{N} I_L + \frac{N_R}{N} I_R$$

Then, we obtain the impurity and impurity reduction to decide the split candidate

$$(X_j^*, s^*) = \underset{X_j, s}{\operatorname{argmin}} I_{split} \Rightarrow \underset{X_j, s}{\operatorname{argmax}} \Delta I$$

# Decision Tree

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## Hyperparameters

**Criterion:** The function to measure the quality of a split, i.e., Gini and entropy.

**Splitter:** The strategy used to choose the split at each node, i.e., best and random.

**max\_depth:** The maximum depth of the tree.

# Decision Tree

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## Hyperparameters

**min\_samples\_split:** The minimum number of samples required to split an internal node.

**min\_samples\_leaf:** The minimum number of samples required to be at a leaf node.

**min\_weight\_fraction\_leaf:** The minimum weighted fraction of the sum total of weights (of all the input samples) required to be at a leaf node.

# Decision Tree

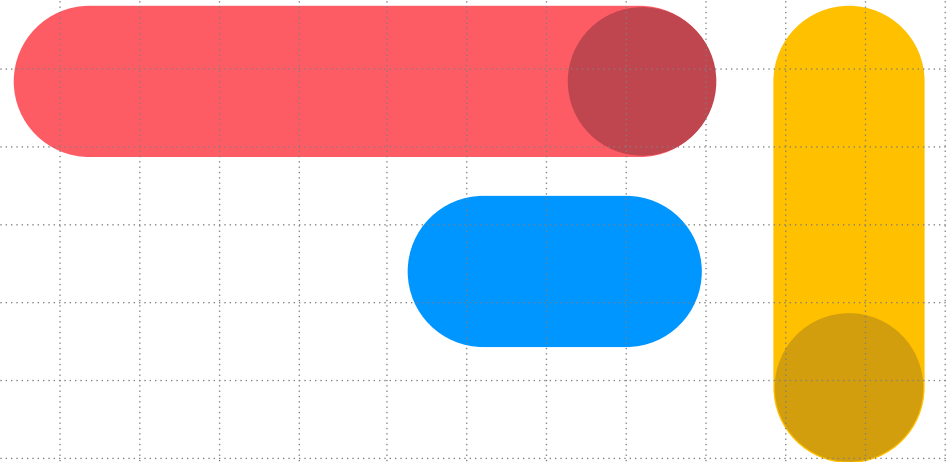
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## Disadvantages

**Decision Tree** has a nature of **Low Bias** and **High Variance**; therefore, if there is a small changes in training dataset, the tree might generate totally different.

**Here come the first question:** How can we reduce the high variance of a single decision tree?

So, instead of relying on one unstable tree, what if we train many trees and combine their predictions?



# The End

**Thank you for your attention!**

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